

Computational Engineering

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Assignment 4: Resolution of the Navier-Stokes equations

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1 Problem Specification

This work addresses the resolution of the Navier-Stokes equations for incompressible flows using the Fractional Step Method (FSM), making use of a structured staggered mesh. Although no particular case will be solved, the general procedure that allows tackling this type of problems through a numerical approach is described in detail.

The physical problem considered consists of anincompressible, unsteady flow with constant viscosity, which can develop in an arbitrary geometry. This type of flow is common in many engineering applications, from lid-driven cavities to flows around bodies or in heated cavities.

The objective of the present work is to explain the numerical procedure that allows solving this system of equations using the fractional step method. For this purpose, the theoretical basis of the method is introduced, its implementation on a staggered mesh, and the necessary steps to carry out the simulation of this type of flows, leaving the resolution of specific cases for future works.

2 Theoretical Background

This section describes the theoretical foundation of the Fractional Step Method (FSM), used to numerically solve the Navier-Stokes equations for incompressible flows. This method allows decoupling the calculation of velocity and pressure through a projection of the velocity field onto a divergence-free space, thus ensuring mass conservation.

2.1 Governing equations

The considered flow is incompressible, unsteady, and with constant physical properties. The equations governing the motion of the fluid are:

- **Continuity equation (incompressibility):**

$$\nabla \cdot \vec{v} = 0$$

- **Momentum equation:**

$$\rho \left(\frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla \vec{v} \right) = -\nabla p + \mu \nabla^2 \vec{v}$$

2.2 Helmholtz-Hodge Theorem

This theorem states that any sufficiently smooth vector field defined in a domain can be decomposed as the sum of a divergence-free field and the gradient of a scalar.

If we take the incompressible Navier-Stokes equations and integrate them in time:

$$\begin{cases} \nabla \cdot \vec{v} = 0 \\ \rho \dfrac{\partial \vec{v}}{\partial t} = \vec{R}(\vec{v}) - \nabla p \end{cases} \tag{1}$$

After time integration:

$$\begin{cases} \nabla \cdot \vec{v}^{n+1} = 0 \\ \rho \dfrac{\vec{v}^{n+1} - \vec{v}^{n}}{\Delta t} = \dfrac{3}{2}\vec{R}(\vec{v}^{n}) - \dfrac{1}{2}\vec{R}(\vec{v}^{n-1}) - \nabla p^{n+1} \end{cases} \tag{2}$$

Applying the Helmholtz-Hodge theorem:

$$\begin{cases} \vec{v}^{n+1} = \vec{v}^{p} - \dfrac{\Delta t}{\rho}\nabla p^{n+1} \\ \rho \dfrac{\vec{v}^{p} - \vec{v}^{n}}{\Delta t} = \dfrac{3}{2}\vec{R}(\vec{v}^{n}) - \dfrac{1}{2}\vec{R}(\vec{v}^{n-1}) \end{cases} \tag{3}$$

By applying the divergence operator and knowing that $\nabla \cdot \vec{v}^{n+1} = 0$, the Poisson equation for pressure is obtained:

$$\Delta p^{n+1} = \frac{\rho}{\Delta t}\nabla \cdot \vec{v}^{p} \tag{4}$$

For each time step, Δp^{n+1} and $\vec{v}^{n+1}$ are unique.

2.3 Checkerboard problem

When computing $\vec{v}^{n+1}$ a problem arises. After applying finite differences, it is observed that Δp^{n+1} at node P is independent of p_p^{n+1}, and although this correctly satisfies the pressure-velocity coupling, it results in unphysical pressure distributions. Therefore, a different coupling strategy is needed.

2.4 FSM for staggered meshes

With staggered meshes, a type of mesh where not all variables are calculated at the same point (P is at the center of the cell, and velocity at the faces), the problem is resolved.

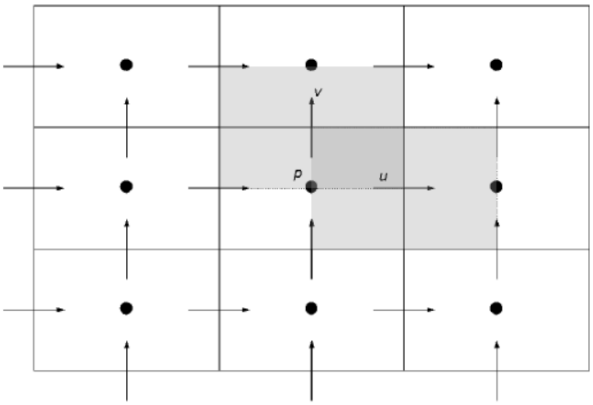


Figure 1: Staggered mesh scheme

2.5 Scheme for solving the INSE

2.5.1 Step 1

The first step to solve the Navier-Stokes equations is to compute the term $R(u) = -(\rho\vec{v} \cdot \nabla)u + \nabla \cdot (\mu\nabla u)$. This term is integrated along the staggered mesh, applying Gauss's theorem and discretizing the result:

$$R(u)\,\Omega_{x_P} = -\left[\dot{m}_e u_e - \dot{m}_w u_w + \dot{m}_n u_n - \dot{m}_s u_s\right] + \left[\mu_e \frac{u_E - u_P}{d_{EP}} A_e - \mu_w \frac{u_P - u_W}{d_{WP}} A_w + \mu_n \frac{u_N - u_P}{d_{NP}} A_n - \mu_s \frac{u_P - u_S}{d_{SP}} A_s\right]$$

The fluxes $\dot{m}_e = (\rho u)_e \cdot A_e$, $\dot{m}_n = (\rho u)_n \cdot A_n$, ... can be calculated using interpolations based on mass conservation. The velocities u_e, u_w, u_n, u_s can be evaluated using numerical schemes (UDS, CDS, ...)

The same process must be repeated in the y direction.

2.5.2 Step 2

In the next step, $\Delta p^{n+1} = \frac{\rho}{\Delta t}\nabla \cdot \vec{v}^p$ is computed. To do this, it is integrated over the main mesh and discretized, yielding the equation:

$$a_P p_P^{n+1} = a_E p_E^{n+1} + a_W p_W^{n+1} + a_N p_N^{n+1} + a_S p_S^{n+1} + b_P \tag{5}$$

The coefficients are:

$$a_P = a_E + a_W + a_N + a_S \tag{6}$$

$$a_E = \frac{A_e}{d_{EP}} \qquad a_N = \frac{A_n}{d_{NP}} \tag{7}$$

$$a_W = \frac{A_w}{d_{WP}} \qquad a_S = \frac{A_s}{d_{SP}} \tag{8}$$

$$b_P = -\frac{1}{\Delta t}\left[(\rho u^p)_e A_e - (\rho u^p)_w A_w + (\rho v^p)_n A_n - (\rho v^p)_s A_s\right] \tag{9}$$

Boundary conditions

In wall-type boundary conditions, where the velocity perpendicular to the surface is zero ($v = 0$), it is assumed that there is no variation of pressure in the normal direction ($\frac{\partial p}{\partial n} = 0$). This allows us to set $v_P^{n+1} = v_P^p$, simplifying the velocity correction. To maintain the boundary condition in the linear system, $a_P = 1$, $a_{nb} = 1$, and the coefficients of the remaining neighbors are set to zero ($a_{i \neq nb} = 0$).

2.5.3 Step 3

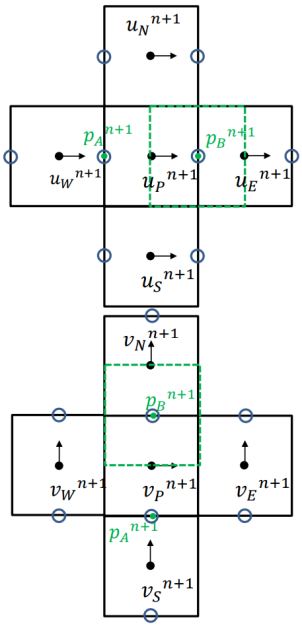


Figure 2: Velocity correction in staggered mesh.

At this step, the corrected velocity components u^{n+1} and v^{n+1} are updated using the *staggered mesh*.

$$u_P^{n+1} = u_P^p - \frac{\Delta t}{\rho} \cdot \frac{p_B^{n+1} - p_A^{n+1}}{d_{BA}}$$

$$v_P^{n+1} = v_P^p - \frac{\Delta t}{\rho} \cdot \frac{p_B^{n+1} - p_A^{n+1}}{d_{BA}}$$

2.5.4 Step 4

Finally, the new timestep is calculated using the Courant-Friedrichs-Lewy stability condition, which ensures that the solution is stable and physically meaningful during time integration.

The value of the timestep must be:

$$\Delta t = \min(\Delta t_c, \Delta t_d)$$

$$\Delta t_c = \min\left(0.35\frac{\Delta x}{|\vec{v}|}\right), \quad \Delta t_d = \min\left(0.20\frac{\Delta x^2}{\mu/\rho}\right)$$

2.5.5 Step 5

This process must be repeated from the initial time to the final time, until the system reaches a steady state.

3 Conclusion

This report has presented the numerical resolution of the incompressible Navier-Stokes equations using the Fractional Step Method. The method has been described in detail, including the decomposition of the velocity field using the Helmholtz-Hodge theorem, the derivation of the pressure Poisson equation, and the time-step strategy.

Although no specific case has been simulated, the procedure provided serves as a foundation for implementing solvers applicable to various incompressible flow problems.